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Power amplifier module

Abstract

A power amplifier module includes a first amplifier that amplifies a power level of a first input signal in a predetermined frequency band and outputs a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier that amplifies a power level of a second input signal in the predetermined frequency band and outputs a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner that combines the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level and includes a transmission line transformer.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8130039	12/2011	Dishop	330/276	H03F 3/211
8593219	12/2012	Root	330/124R	H03F 1/0288
2011/0309884	12/2010	Dishop	N/A	N/A
2013/0099866	12/2012	Conradi	330/126	H03F 1/0288
2016/0191105	12/2015	King et al.	N/A	N/A
2017/0104456	12/2016	Motoi et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2016042699	12/2015	JP	N/A
2015133003	12/2014	WO	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority from Japanese Patent Application No. 2020-122668 filed on Jul. 17, 2020. The content of this application is incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

(2) The present disclosure relates to a power amplifier module.

2. Description of the Related Art

(3) In recent years, in mobile terminals, for a radio frequency (RF) front-end circuit, a device supporting multiple frequency bands specified in the third generation partnership project (3GPP) is used. Furthermore, high-speed communications are demanded, and multiband operation using multiple frequency bands simultaneously is therefore adopted. Thus, a technique is disclosed in which signals amplified by respective amplifiers of two systems are combined to generate a signal compliant with an intended communication standard (see Japanese Unexamined Patent Application Publication No. 2016-42699).

(4) In a front-end circuit disclosed in Japanese Unexamined Patent Application Publication No. 2016-42699, 3G/4G signals of the two systems are combined by a power combiner including a matching network and a band selection switch, and then a 2G signal is outputted through an impedance transformation circuit. However, in the front-end circuit, an LC ladder circuit is used for the matching network, and thus a wider bandwidth is not able to be achieved.

(5) Furthermore, International Publication No. WO2015/133003A1 discloses an amplifier circuit that combines amplified signals. The amplifier circuit disclosed in International Publication No. WO2015/133003A1 combines signals through respective quarter-wave lines provided in two respective systems to generate an output signal. However, in the amplifier circuit disclosed in International Publication No. WO2015/133003A1, output matching is performed by using quarter-wave lines whose optimum line lengths differ according to a frequency, and thus it is difficult to achieve a wider bandwidth. In other words, in the amplifier circuit, wideband signals are not able to be combined. Furthermore, the area occupied by each quarter-wave line in the circuit is large, thus resulting in an increase in module size.

BRIEF SUMMARY OF THE DISCLOSURE

(6) Thus, the present disclosure has been made to appropriately combine wideband signals while achieving a reduction in module size.

(7) A power amplifier module according to one aspect of the present disclosure includes a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level and including a transmission line transformer.

(8) The present disclosure can provide the power amplifier module that can appropriately combine wideband signals while achieving a reduction in module size. Other features, elements, characteristics and advantages of the present disclosure will become more apparent from the following detailed description of preferred embodiments of the present disclosure with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 illustrates an overview of a configuration of a power amplifier module according to a present embodiment;

(2) FIG. 2 illustrates an example of the configuration of the power amplifier module according to the present embodiment;

(3) FIG. 3A schematically illustrates an example of a transmission line transformer;

(4) FIG. 3B is a perspective view schematically illustrating an example of a structure of the

transmission line transformer;

(5) FIG. 3C is a perspective view schematically illustrating an example of the structure of the transmission line transformer;

(6) FIG. 4 illustrates an example of a configuration of a first form of combiner;

(7) FIG. 5 illustrates an example of a configuration of a second form of combiner;

(8) FIG. 6 illustrates an example of a configuration of a third form of combiner;

(9) FIG. 7 illustrates an example of a configuration of a fourth form of combiner;

(10) FIG. 8 illustrates an example of a configuration of a fifth form of combiner;

(11) FIG. 9A illustrates an overview of a configuration of a power amplifier module according to another embodiment;

(12) FIG. 9B illustrates an overview of a configuration of the power amplifier module according to the other embodiment; and

(13) FIG. 10 illustrates an overview of a configuration of a power amplifier module according to a comparative example.

DETAILED DESCRIPTION OF THE DISCLOSURE

(14) Embodiments of the present disclosure will be described below with reference to the drawings. Here, circuit elements denoted by the same reference numerals are the same circuit element, and repeated descriptions of the circuit elements are omitted.

(15) Configuration of Power Amplifier Module **100**

(16) An overview of a power amplifier module **100** according to a present embodiment will be described with reference to FIG. 1. FIG. 1 illustrates an overview of a configuration of the power amplifier module **100** according to the present embodiment. The power amplifier module **100** is incorporated, for example, in a mobile communication device, such as a cellular phone. The power amplifier module **100** amplifies the power of an input signal RFin to a level necessary for transmission to a base station and outputs this input signal RFin as an amplified signal RFout. The input signal RFin is a radio frequency (RF) signal modulated, for example, by a radio frequency integrated circuit (RFIC) or the like in accordance with a certain communication scheme. Examples of a communication standard of the input signal RFin include the second generation mobile communication system (2G), the third generation mobile communication system (3G), the fourth generation mobile communication system (4G), the fifth generation mobile communication system (5G), 5G new radio (5G NR), long term evolution (LTE)-frequency division duplex (FDD), LTE-time division duplex (TDD), LTE-Advanced, and LTE-Advanced Pro. The frequency of the input signal RFin ranges, for example, from about several hundred MHz to about several tens of GHz. The communication standard and frequency of the input signal RFin are not limited to these.

(17) For example, between a mobile terminal using downlink multiple-input and multiple-output (MIMO), such as a mobile terminal supporting 5G NR, and a base station, a reference signal for grasping a radio wave propagation situation is transmitted before data communication. At this time, a reference signal is outputted to two antennas of a plurality of antennas incorporated in the mobile terminal. On the other hand, when an uplink is established from the mobile terminal to the base station, a PC2 transmission from one antenna and PC3 transmissions from two respective antennas are performed.

(18) Here, each of PC2 and PC3 denotes "Power Class" representing an output level of a signal and is specified in 3GPP. With respect to PC3, for example, transmission power transmitted from an antenna of the mobile terminal is about 23 dBm. With respect to PC2, for example, transmission power transmitted from an antenna of the mobile terminal is about 26 dBm. In other words, PC2 is higher than PC3 in transmission power by about 3 dBm, and thus the output of PC2 is about twice the output of PC3.

(19) Hence, in mobile terminals supporting 5G NR, for example, an aspect in which a reference signal is transmitted, an aspect in which a PC2 transmission from one antenna is performed, and an aspect in which PC3 transmissions from two respective antennas are performed have to be

implemented. For at least these aspects, the power amplifier module **100** according to the present embodiment can achieve high-efficiency transmission.

(20) First, an overview of the above-described aspects in the power amplifier module **100** will be described with reference to FIG. **1**. In FIG. **1**, as an example, communication paths that transmit PC3 signals from two respective antennas ANT1 and ANT2 are indicated by a dashed line, and a communication path that transmits a PC2 signal from one antenna ANT1 (or ANT2) is indicated by a dash-dot-dot line.

(21) As illustrated in FIG. **1**, an input signal RFin1 is inputted from a terminal TX1 to a first amplifier **110a**, and an input signal RFin2 is inputted from a terminal TX2 to a second amplifier **110b**. The first amplifier **110a** and the second amplifier **110b** are optimally designed, for example, for PC3.

(22) In the power amplifier module **100**, when PC3 signals are transmitted from the two respective antennas ANT1 and ANT2, an input signal RFin1 is transmitted to the antenna ANT1 through the first amplifier **110a**, a first band selection switch **140a**, a duplexer **150**, and an antenna switch **160**. Similarly, an input signal RFin2 is transmitted to the antenna ANT2 through the second amplifier **110b**, a second band selection switch **140b**, a duplexer **150**, and the antenna switch **160**. Although not illustrated in FIG. **1**, behind the first and second band selection switches **140a** and **140b**, a plurality of duplexers **150** suited to respective bands may be connected.

(23) On the other hand, in the power amplifier module **100**, when a PC2 signal is transmitted from one antenna ANT2, an input signal RFin1 is inputted to a combiner **180** through the first amplifier **110a**, the first band selection switch **140a**, the duplexer **150**, and the antenna switch **160**. An input signal RFin2 is inputted to the combiner **180** through the second amplifier **110b**, the second band selection switch **140b**, the duplexer **150**, and the antenna switch **160**. The combiner **180** combines the input signal RFin1 of PC3 (amplified signal RFamp1) and the input signal RFin2 of PC3 (amplified signal RFamp2) and outputs a PC2 signal. Here, the combiner **180** includes a transmission line transformer. The signal outputted from the combiner **180** is transmitted to the antenna ANT2 through a path-changing switch **170**.

(24) In other words, the power amplifier module **100** can transmit PC3 signals to the respective antennas ANT1 and ANT2 of two systems by using two amplifiers optimally designed for PC3 and can further transmit a PC2 signal to the antenna ANT1 (or ANT2) of one system.

(25) Here, a power amplifier module **1000** according to a comparative example will be described with reference to FIG. **10**. FIG. **10** illustrates an overview of a configuration of the power amplifier module **1000** according to the comparative example. In FIG. **10**, as an example, communication paths that transmit PC3 signals from the two respective antennas ANT1 and ANT2 are indicated by a dashed line, and a communication path that transmits a PC2 signal from one antenna ANT1 is indicated by a dash-dot-dot line.

(26) In the power amplifier module **1000**, when PC3 signals are transmitted from the two respective antennas ANT1 and ANT2, an input signal RFin1 is inputted to an amplifier **1112** provided in or on a given substrate S1, and an input signal RFin2 is inputted to an amplifier **1210** provided in or on a substrate S2 different from the given substrate S1. Here, each of the amplifiers **1112** and **1210** is optimally designed for PC2 to implement a PC2 transmission using one antenna. The input signals RFin1 and RFin2 are transmitted to the respective antennas ANT1 and ANT2 through band selection switches **1142** and **1240** and duplexers **1152** and **1250** that are provided in or on the different respective substrates S1 and S2, and through an antenna switch **1160** provided in or on the given substrate S1.

(27) As described above, in comparison with the power amplifier module **1000** according to the comparative example, in the power amplifier module **100** according to the present embodiment, for example, the first amplifier **110a** and the second amplifier **110b** are optimally designed for PC3, thus enabling an increase in transmission efficiency.

(28) Furthermore, in the power amplifier module **100**, for example, PC3 signals are combined to

generate a PC2 signal, and a component (substrate S2 of the power amplifier module **1000**) for generating a PC2 signal is therefore unnecessary, enabling a reduction in the number of components and a reduction in package size. Additionally, since the combiner **180** includes the transmission line transformer, a wide frequency band including a middle band (MB) and a high band (HB) can be supported by one combiner **180**. Furthermore, the combiner **180** is disposed on an antennas ANT1 and ANT2 side of the antenna switch **160**, and thus impedances at input and output terminals of the antenna switch **160** are about 50 ohms, therefore making it possible to reduce power loss due to a resistance component of the antenna switch **160**.

(29) Next, the configuration of the power amplifier module **100** will be described in detail with reference to FIG. 2. FIG. 2 illustrates an example of the configuration of the power amplifier module **100** according to the present embodiment.

(30) The power amplifier module **100** includes, for example, the first amplifier **110a**, a first impedance transformer **120a**, a first filter circuit **130a**, the first band selection switch **140a**, the second amplifier **110b**, a second impedance transformer **120b**, a second filter circuit **130b**, the second band selection switch **140b**, duplexers **150**, the antenna switch **160**, the path-changing switch **170**, and the combiner **180**. Each component will be described in detail below.

(31) The first amplifier **110a** is, for example, a circuit that amplifies a power level of an input signal RFin1 and outputs an amplified signal RFamp1. The first amplifier **110a** supports, for example, input signals RFin1 in a frequency band (about 1710 MHz to about 2690 MHz) including the MB and the HB. The first amplifier **110a** is optimally designed, for example, for PC3.

(32) The first impedance transformer **120a** is, for example, a circuit that performs impedance transformation at a predetermined transformation ratio, and includes a transmission line transformer. In FIG. 2, as an example, a transformation ratio of "1:9 to 25" is indicated. When the transmission line transformer is included, wideband and low-loss impedance transformation can be achieved. A transmission line transformer will be described in detail below.

(33) The first filter circuit **130a** subjects, for example, a signal in the frequency band including the MB and the HB to filtering. The first filter circuit **130a** may be, for example, a fixed filter whose pass band is fixed at a specific frequency band, a variable filter whose pass band is varied in accordance with each of multiple frequency bands, or a filter obtained by combining the fixed filter and the variable filter such that they are switchable. Thus, attenuation characteristics in proximity to the pass band can be increased.

(34) The first band selection switch (BSSW) **140a** is, for example, a switch that distributes a plurality of high-frequency signals whose frequency bands are different to intended terminals. In the first band selection switch **140a**, the amplified signal RFamp1 is inputted to an input terminal, and this signal is outputted from an intended output terminal.

(35) The second amplifier **110b**, the second impedance transformer **120b**, the second filter circuit **130b**, and the second band selection switch **140b** are similar to the first amplifier **110a**, the first impedance transformer **120a**, the first filter circuit **130a**, and the first band selection switch **140a**, and thus a description thereof is omitted.

(36) In other words, high-frequency signals being substantially equal in phase and amplitude are inputted to the first amplifier **110a** and the second amplifier **110b**. Then, the two amplifiers **110a** and **110b** amplify the input high-frequency signals and output the high-frequency signals being substantially equal in phase and amplitude. The high-frequency signals amplified by the two amplifiers **110a** and **110b** are subjected to impedance transformation in transmission line transformers in the respective impedance transformers **120a** and **120b** and then are outputted to duplexers **150**.

(37) Each of the duplexers (DPXs) **150** is, for example, a filter circuit that, when a transmission frequency and a reception frequency are different, switches between transmission and reception. The duplexers **150** are connected, for example, between output terminals of the first and second band selection switches **140a** and **140b** and input terminals **161** of the antenna switch **160** to be

described. FIG. 2 illustrates distribution of an amplified signal RFamp1 and a reception signal RX by a duplexer 150. Incidentally, if a communication scheme is, for example, a time division duplex (TDD) scheme, the duplexers 150 may be omitted. Instead, the duplexer 150 may be replaced by typical band pass filters.

(38) The antenna switch (ANTSW) 160 is, for example, a switch that switches between a path connected to each of the antennas ANT1 and ANT2 and a path connected to the combiner 180 to be described. Thus, it can be determined selectively whether to output PC3 signals to the respective antennas ANT1 and ANT2 or to input two PC3 signals to the combiner 180 to output a PC2 signal.

(39) Specifically, the antenna switch 160 includes, for example, the input terminals 161, a first output terminal 162a, a second output terminal 162b, and third output terminals 162c. The input terminals 161 are, for example, terminals that are connected to the respective duplexers 150 and to which an amplified signal RFamp1 and an amplified signal RFamp2 are inputted. The first output terminal 162a is, for example, a terminal connected to the antenna ANT1 so as to output the amplified signal RFamp1 to the antenna ANT1. The second output terminal 162b is, for example, a terminal connected to the antenna ANT2 so as to output the amplified signal RFamp2 to the antenna ANT2. The third output terminals 162c are, for example, terminals connected to the combiner 180 so as to output the amplified signal RFamp1 and the amplified signal RFamp2 to the combiner 180.

(40) The path-changing switch 170 is a switch that distributes a signal outputted from the combiner 180 to be described to an intended antenna.

(41) The combiner 180 is, for example, a circuit that combines PC3 signals that are the amplified signals RFamp1 and RFamp2 to generate a PC2 signal. The combiner 180 is connected between the antenna switch 160 and the antennas ANT1 and ANT2. Thus, impedances at all input and output terminals of the antenna switch 160 are substantially 50 ohms, therefore making it possible to reduce power loss due to a resistance component of the antenna switch 160. The combiner 180 is constructed, for example, by combining transmission line transformers. When the combiner 180 includes the transmission line transformers, wideband and low-loss impedance transformation can be achieved.

(42) First, a structure of a transmission line transformer 190 will be described below with reference to FIGS. 3A and 3B. Then, variations of the combiner 180 obtained by combining transmission line transformers 190 will be described with reference to FIGS. 4 to 8.

(43) FIG. 3A schematically illustrates an example of the transmission line transformer 190. FIGS. 3B and 3C are perspective views schematically illustrating an example of the structure of the transmission line transformer 190.

(44) As illustrated in FIG. 3A, the transmission line transformer 190 includes, for example, a first transmission line 191, a second transmission line 192, and a third transmission line 193. First, alternating currents (AC) that flow through the first transmission line 191, the second transmission line 192, and the third transmission line 193 will be described. A current that flows from an input terminal 190a toward an output terminal 190b flows first from, of the first transmission line 191, a first end portion 191a toward a second end portion 191b. Then, the current flows from a third end portion 192a toward a fourth end portion 192b in the second transmission line 192. The magnitude of the AC current that flows through the first transmission line 191 is equal to the magnitude of the AC current that flows through the second transmission line 192. In the third transmission line 193, an odd mode current is induced from a fifth end portion 193a toward a sixth end portion 193b by the AC current that flows through the first transmission line 191. In the third transmission line 193, an odd mode current is induced from the fifth end portion 193a toward the sixth end portion 193b by the AC current that flows through the second transmission line 192. A direction of an odd mode current induced in the third transmission line 193 is opposite to directions of the AC currents that flow through the first transmission line 191 and the second transmission line 192. The odd mode current caused by the current that flows through the first transmission line 191 and the odd mode

current caused by the current that flows through the second transmission line **192** are equal in magnitude and direction.

(45) In other words, in the third transmission line **193**, the odd mode current derived from the first transmission line **191** and the odd mode current derived from the second transmission line **192** are superimposed and flow. For this reason, in the third transmission line **193**, an odd mode current is induced whose magnitude is twice the magnitude of a current that flows through a series circuit composed of the first transmission line **191** and the second transmission line **192**. When the magnitude of a current that flows from the input terminal **190a** into the transmission line transformer **190** is denoted by i , a current of $(\frac{1}{3})i$ flows through the series circuit composed of the first transmission line **191** and the second transmission line **192**, and a current of $(\frac{2}{3})i$ flows through the third transmission line **193**. Hence, the magnitude of the current outputted from the output terminal **190b** is $(\frac{1}{3})i$.

(46) Next, a voltage will be described. A voltage at the input terminal **190a** is denoted by v_1 , and a voltage at the output terminal **190b** is denoted by v_2 . A voltage at the first end portion **191a** of the first transmission line **191** and a voltage at the fifth end portion **193a** of the third transmission line **193** are equal to the voltage v_1 at the input terminal **190a**. A voltage at the fourth end portion **192b** of the second transmission line **192** is equal to the voltage v_2 at the output terminal **190b**. A voltage at the second end portion **191b** of the first transmission line **191** is denoted by v_3 . A voltage at the third end portion **192a** of the second transmission line **192** is equal to the voltage v_3 at the second end portion **191b** of the first transmission line **191**. A voltage at the sixth end portion **193b** of the third transmission line **193** is 0 V.

(47) A potential difference between the first end portion **191a** and the second end portion **191b** of the first transmission line **191** is equal to a potential difference between the sixth end portion **193b** and the fifth end portion **193a** of the third transmission line **193**, and thus an equation of $v_1 - v_3 = 0 - v_1$ holds. Similarly, for a relationship between the second transmission line **192** and the third transmission line **193**, an equation of $v_3 - v_2 = 0 - v_1$ holds. When the simultaneous equations are solved, $3 \times v_1 = v_2$ is obtained. Thus, the voltage v_2 at the output terminal **190b** is three times the voltage v_1 at the input terminal **190a**.

(48) When a load having an impedance R_2 is connected to the output terminal **190b**, an equation of $v_2 = (\frac{1}{3})i \times R_2$ holds. When an impedance as seen from the input terminal **190a** looking toward a load side is denoted by R_1 , an equation $v_1 = R_1 \times i$ holds. When these equations are solved, $R_1 = (\frac{1}{9})R_2$ is obtained. Thus, the impedance R_1 as seen from the input terminal **190a** looking toward the load side is 1/9 times the impedance R_2 of the load connected to the output terminal **190b**. On the other hand, when the load is connected to the input terminal **190a**, an impedance as seen from the output terminal **190b** looking toward a load side is nine times an impedance of the load connected to the input terminal **190a**. Thus, the transmission line transformer **190** according to the present embodiment functions as an impedance transformation circuit with an impedance transformation ratio of nine.

(49) As illustrated in FIG. 3B, in the transmission line transformer, the first to third transmission lines are disposed so as to be stacked in a thickness direction of a substrate. Specifically, the first end portion **191a** of the first transmission line **191**, the fifth end portion **193a** of the third transmission line **193**, and the fourth end portion **192b** of the second transmission line **192** are disposed such that they overlay one another when viewed in plan. The second end portion **191b** of the first transmission line **191** and the third end portion **192a** of the second transmission line **192** are disposed such that they overlay each other when viewed in plan. In the same layer where the third transmission line **193** is disposed, a conductor pattern **199** is disposed at a position corresponding to the second end portion **191b** of the first transmission line **191**. A via conductor **195** connects the first end portion **191a** of the first transmission line **191** and the fifth end portion **193a** of the third transmission line **193**. A via conductor **196** connects the second end portion **191b** of the first transmission line **191** and the conductor pattern **199**, and a via conductor **197** connects

the conductor pattern **199** and the third end portion **192a** of the second transmission line **192**. A via conductor **198** connects the fourth end portion **192b** of the second transmission line **192** and an extended line **194**. The first end portion **191a** of the first transmission line **191** is connected to the input terminal **190a**, and the extended line **194** is connected to the output terminal **190b**. The sixth end portion **193b** of the third transmission line **193** is connected to a ground conductor GND.

(50) As described above, the transmission line transformer **190** illustrated in FIGS. 3A and 3B has an impedance transformation ratio of 1:9, whereas, as illustrated in FIG. 3C, the transmission line transformer **190**, for example, with a two-layer structure composed of the first transmission line **191** and the third transmission line **193** has an impedance transformation ratio of 1:4. Furthermore, in the transmission line transformer **190** illustrated in FIGS. 3A and 3B, an impedance transformation ratio of 1:2.25 is provided by connecting the first end portion **191a** of the first transmission line **191** to the fifth end portion **193a** of the third transmission line **193**, connecting the sixth end portion **193b** of the third transmission line **193** to the fourth end portion **192b** of the second transmission line **192**, and connecting the third end portion **192a** of the second transmission line **192** to the ground conductor GND.

(51) In the present embodiment, there will be described, as an example, the combiner **180** constructed by any combination of a transmission line transformer with an impedance transformation ratio of 1:9 (or 9:1), a transmission line transformer with a transformation ratio of 1:4 (or 4:1), and a transmission line transformer with a transformation ratio of 1:2.25 (or 2.25:1). Note that the combiner **180** is constructed by combining transmission line transformers with various transformation ratios. For example, the combiner **180** only has to be constructed so that an impedance at an input terminal is substantially equal to an impedance at an output terminal. For convenience of explanation, the following description will be given assuming that a portion corresponding to an end portion connected to the input terminal **190a** in the transmission line transformer **190** is referred to as “one end” and a portion corresponding to an end portion connected to the output terminal **190b** is referred to as “the other end”.

(52) First Form of Combiner **180a**

(53) A first form of combiner **180a** will be described with reference to FIG. 4. FIG. 4 illustrates an example of a configuration of the first form of combiner **180a**.

(54) As illustrated in FIG. 4, the combiner **180a** includes, for example, a first input terminal **181a**, a first transmission line transformer **182a**, a second input terminal **183a**, a second transmission line transformer **184a**, a first isolation section **185a**, a combination terminal **186a**, a third transmission line transformer **187a**, and an output terminal **188a**.

(55) The first input terminal **181a** is a terminal to which an amplified signal RFamp1 is inputted. The first transmission line transformer **182a** is, for example, a circuit that performs impedance transformation at a transformation ratio of 1:9. In the first transmission line transformer **182a**, one end is connected to the first input terminal **181a**, and the other end is connected to the combination terminal **186a**. The second input terminal **183a** is a terminal to which an amplified signal RFamp2 is inputted. The second transmission line transformer **184a** is, for example, a circuit that performs impedance transformation at the same transformation ratio of 1:9 as that of the first transmission line transformer **182a**. In the second transmission line transformer **184a**, one end is connected to the second input terminal **183a**, and the other end is connected to the combination terminal **186a**. In other words, the first transmission line transformer **182a** and the second transmission line transformer **184a** are connected in parallel. The first isolation section **185a** includes a resistor R and a capacitor C connected in parallel. The first isolation section **185a** is connected between the first input terminal **181a** and the second input terminal **183a**. The combination terminal **186a** is a terminal that combines the amplified signal RFamp1 inputted through the first transmission line transformer **182a** and the amplified signal RFamp2 inputted through the second transmission line transformer **184a**. The third transmission line transformer **187a** is, for example, a circuit that performs impedance transformation at a transformation ratio of 4:1. In the third transmission line

transformer **187a**, one end is connected to the combination terminal **186a**, and the other end is connected to the output terminal **188a**. The output terminal **188a** is connected to the path-changing switch **170**.

(56) Next, an overview of impedance transformation performed in the combiner **180a** will be described. An impedance at the first input terminal **181a** of the combiner **180a** is Z_S (50 ohms). In the first transmission line transformer **182a**, Z_S (50 ohms) is transformed into $9 \times Z_S$ (450 ohms). The same things hold true for the second input terminal **183a** and the second transmission line transformer **184a**. Then, at the combination terminal **186a**, an impedance of $9 \times Z_S \times \frac{1}{2}$ (225 ohms) is reached and is transformed into a quarter in the third transmission line transformer **187a**. Hence, an impedance (Z_L) at the output terminal **188a** is $Z_S \times \frac{9}{8}$ (about 50 ohms). Incidentally, in the third transmission line transformer **187a**, the impedance may be finely adjusted.

(57) Thus, in comparison with the power amplifier module **1000** according to the comparative example, the low-loss and wideband power amplifier module **100** can be implemented.

(58) Second Form of Combiner **180b**

(59) A second form of combiner **180b** will be described with reference to FIG. 5. FIG. 5 illustrates an example of a configuration of the second form of combiner **180b**.

(60) As illustrated in FIG. 5, the combiner **180b** includes, for example, a third input terminal **181b**, a fourth transmission line transformer **182b**, a fourth input terminal **183b**, a fifth transmission line transformer **184b**, a second isolation section **185b**, a combination terminal **186b**, a sixth transmission line transformer **187b**, and an output terminal **188b**.

(61) The second form of combiner **180b** is a combiner in which, in the first form of combiner **180a**, the first and second transmission line transformers **182a** and **184a** with the transformation ratio of 1:9 are changed to the fourth and fifth transmission line transformers **182b** and **184b** with a transformation ratio of 4:1 and the third transmission line transformer **187a** with the transformation ratio of 4:1 is changed to the sixth transmission line transformer **187b** with a transformation ratio of 1:9. This enables low-loss and wideband power combination.

(62) Third Form of Combiner **180c**

(63) A third form of combiner **180c** will be described with reference to FIG. 6. FIG. 6 illustrates an example of a configuration of the third form of combiner **180c**.

(64) As illustrated in FIG. 6, the combiner **180c** includes, for example, a fifth input terminal **181c**, a seventh transmission line transformer **182c**, an eighth transmission line transformer **183c**, a sixth input terminal **184c**, a ninth transmission line transformer **185c**, a tenth transmission line transformer **186c**, a third isolation section **187c**, a combination terminal **188c**, a first connection terminal **1881c**, a second connection terminal **1882c**, and an output terminal **189c**.

(65) The fifth input terminal **181c** is a terminal to which an amplified signal RFamp1 is inputted. The seventh transmission line transformer **182c** is, for example, a circuit that performs impedance transformation at a transformation ratio of 1:9. In the seventh transmission line transformer **182c**, one end is connected to the fifth input terminal **181c**, and the other end is connected to one end of the eighth transmission line transformer **183c** through the first connection terminal **1881c**. The eighth transmission line transformer **183c** is, for example, a circuit that performs impedance transformation at a transformation ratio of 4:1. In the eighth transmission line transformer **183c**, the other end is connected to the combination terminal **188c**. The sixth input terminal **184c** is a terminal to which an amplified signal RFamp2 is inputted. The ninth transmission line transformer **185c** is, for example, a circuit that performs impedance transformation at the same transformation ratio of 1:9 as that of the seventh transmission line transformer **182c**. In the ninth transmission line transformer **185c**, one end is connected to the sixth input terminal **184c**, and the other end is connected to one end of the tenth transmission line transformer **186c** through the second connection terminal **1882c**. The tenth transmission line transformer **186c** is, for example, a circuit that performs impedance transformation at the same transformation ratio of 4:1 as that of the eighth transmission line transformer **183c**. In the tenth transmission line transformer **186c**, the other end is

connected to the combination terminal **188c**. In other words, the seventh transmission line transformer **182c** and the eighth transmission line transformer **183c** are connected in parallel with the ninth transmission line transformer **185c** and the tenth transmission line transformer **186c**. The third isolation section **187c** is connected between the fifth input terminal **181c** and the sixth input terminal **184c**. The third isolation section **187c** is, for example, a resistor having a resistance value based on a power distribution ratio between a transmission line connected to the fifth input terminal **181c** and a transmission line connected to the sixth input terminal **184c**. The combination terminal **188c** is a terminal that combines the amplified signal RFamp1 inputted through the seventh transmission line transformer **182c** and the eighth transmission line transformer **183c** and the amplified signal RFamp2 inputted through the ninth transmission line transformer **185c** and the tenth transmission line transformer **186c**. The output terminal **189c** is connected to the combination terminal **188c**. The output terminal **189c** is connected to the path-changing switch **170**.

(66) Next, an overview of impedance transformation performed in the combiner **180c** will be described. An impedance at the fifth input terminal **181c** of the combiner **180c** is Z_S (50 ohms). In the seventh transmission line transformer **182c**, Z_S (50 ohms) is transformed into $9 \times Z_S$ (450 ohms). Then, $9 \times Z_S$ (450 ohms) is transformed into $Z_S \times 9/4$ (112 ohms) in the eighth transmission line transformer **183c**. The same things hold true for the sixth input terminal **184c**, the ninth transmission line transformer **185c**, and the tenth transmission line transformer **186c**. Then, at the combination terminal **188c**, $Z_S \times 9/4$ (112 ohms) is reduced by half. Hence, an impedance at the output terminal **189c** is $Z_S \times 9/8$ (about 50 ohms). Incidentally, in the eighth transmission line transformer **183c** and the tenth transmission line transformer **186c**, impedances may be finely adjusted.

(67) Thus, in comparison with the power amplifier module **1000** according to the comparative example, the low-loss and wideband power amplifier module **100** can be implemented.

(68) Fourth Form of Combiner **180d**

(69) A fourth form of combiner **180d** will be described with reference to FIG. 7. FIG. 7 illustrates an example of a configuration of the fourth form of combiner **180d**.

(70) As illustrated in FIG. 7, the fourth form of combiner **180d** includes, for example, a seventh input terminal **181d**, an eleventh transmission line transformer **182d**, a twelfth transmission line transformer **183d**, an eighth input terminal **184d**, a thirteenth transmission line transformer **185d**, a fourteenth transmission line transformer **186d**, a fourth isolation section **187d**, a combination terminal **188d**, and an output terminal **189d**. The fourth form of combiner **180d** is a combiner in which, in the third form of combiner **180c**, the seventh and ninth transmission line transformers **182c** and **185c** with the transformation ratio of 1:9 are changed to the eleventh and thirteenth transmission line transformers **182d** and **185d** with a transformation ratio of 4:1 and the eighth and tenth transmission line transformers **183c** and **186c** with the transformation ratio of 4:1 are changed to the twelfth and fourteenth transmission line transformers **183d** and **186d** with a transformation ratio of 1:9. This enables low-loss and wideband power combination.

(71) Fifth Form of Combiner **180e**

(72) A fifth form of combiner **180e** will be described with reference to FIG. 8. FIG. 8 illustrates an example of a configuration of the fifth form of combiner **180e**.

(73) As illustrated in FIG. 8, the combiner **180e** includes, for example, a ninth input terminal **181e**, a tenth input terminal **182e**, a fifteenth transmission line transformer **183e**, a sixteenth transmission line transformer **184e**, a fifth isolation section **185e**, an output terminal **186e**, and a combination terminal **187e**.

(74) The ninth input terminal **181e** is a terminal to which an amplified signal RFamp1 is inputted. The tenth input terminal **182e** is a terminal to which an amplified signal RFamp2 is inputted. The fifteenth transmission line transformer **183e** is, for example, a circuit that performs impedance transformation at a transformation ratio of 4:1. In the fifteenth transmission line transformer **183e**, for example, the ninth input terminal **181e** is connected to the second end portion **191b** of the first

transmission line **191** of the transmission line transformer **190** illustrated in FIG. 3C, the tenth input terminal **182e** is connected to the sixth end portion **193b** (second end portion) of the third transmission line **193**, and the combination terminal **187e** is connected to the first end portion **191a** of the first transmission line **191**. Here, although FIG. 3C illustrates the transmission line transformer **190** including the extended line **194**, the transmission line transformer **190** does not have to include the extended line **194**. In this case, the second end portion **191b** of the first transmission line **191** is constructed so as to extend toward the outside of the first transmission line **191**, and the ninth input terminal **181e** may be provided at the extended second end portion **191b**. In the fifteenth transmission line transformer **183e**, the other end is connected to one end of the sixteenth transmission line transformer **184e** through the combination terminal **187e**. The sixteenth transmission line transformer **184e** is, for example, a circuit that performs impedance transformation at a transformation ratio of 1:2.25. In the sixteenth transmission line transformer **184e**, the one end is connected to the other end of the fifteenth transmission line transformer **183e**, and the other end is connected to the output terminal **186e**. The output terminal **186e** is connected to the path-changing switch **170**.

(75) Next, an overview of impedance transformation performed in the combiner **180e** will be described. Each of impedances at the ninth and tenth input terminals **181e** and **182e** of the combiner **180e** is Z_S (50 ohms). Here, the fifteenth transmission line transformer **183e** performs an operation of transforming the sum of impedances connected between the ninth input terminal **181e** and the tenth input terminal **182e** into a quarter. When powers in phase are inputted to the respective ninth and tenth input terminals **181e** and **182e** of the combiner **180e**, an impedance between the ninth input terminal **181e** and the tenth input terminal **182e** can be regarded as a configuration in which two Z_S s are connected in series through a common terminal (ground) and is thus 100 ohms from $2 \times Z_S$. In the fifteenth transmission line transformer **183e**, $2 \times Z_S$ (100 ohms) is transformed into $2 \times Z_S \times \frac{1}{4}$ (25 ohms), and 25 ohms is reached at the combination terminal **187e** (combination terminal). Then, 25 ohms is transformed into $Z_S/2 \times 2.25$ in the sixteenth transmission line transformer **184e**, and an impedance at the output terminal **186e** is $Z_S/2 \times 2.25$ (about 50 ohms). Incidentally, in the sixteenth transmission line transformer **184e**, the impedance may be finely adjusted.

(76) Thus, in comparison with the power amplifier module **1000** according to the comparative example, the low-loss and wideband power amplifier module **100** can be implemented.

(77) Other Embodiments

(78) A power amplifier module **200** according to another embodiment will be described with reference to FIGS. 9A and 9B. FIGS. 9A and 9B illustrate an overview of a configuration of the power amplifier module **200** according to the other embodiment.

(79) The power amplifier module **200** according to the other embodiment includes a combiner **280** including the antenna switch **160** of the power amplifier module **100** illustrated in FIG. 1. Other components are similar to those of the power amplifier module **100**, and thus a description thereof is omitted.

(80) FIG. 9A illustrates the power amplifier module **200** including a transmission line transformer between an antenna switch **282** and the antennas ANT1 and ANT2. As illustrated in FIG. 9A, the combiner **280** includes, for example, preceding transmission line transformers **281**, the antenna switch **282**, a subsequent transmission line transformer **283**, and a combination terminal **284**.

(81) The preceding transmission line transformers **281** are connected to antennas ANT1 and ANT2 sides of duplexers **250**. Each of the preceding transmission line transformers **281** performs impedance transformation at a predetermined transformation ratio. Specifically, the preceding transmission line transformers **281** correspond to, for example, in the power amplifier module **100**, the first and second transmission line transformers **182a** and **184a** in the combiner **180a**, or the fourth and fifth transmission line transformers **182b** and **184b** in the combiner **180b**.

(82) Although the input terminals of the antenna switch **160** illustrated in FIG. 2 are connected to

the duplexers **150**, input terminals of the antenna switch **282** are connected, for example, to the preceding transmission line transformers **281** in place of the duplexers **150**. The structure of the antenna switch **282** is similar to the structure of the antenna switch **160** illustrated in FIG. 2, and thus a description thereof is omitted. In comparison with a combination terminal provided in the combiner **180** of the power amplifier module **100**, in the power amplifier module **200** according to the other embodiment, the combination terminal **284** is provided, for example, between the antenna switch **282** and the subsequent transmission line transformer **283**.

(83) The subsequent transmission line transformer **283** is connected to an antennas ANT1 and ANT2 side of the antenna switch **282**. The subsequent transmission line transformer **283** performs impedance transformation at a predetermined transformation ratio. Specifically, the subsequent transmission line transformer **283** corresponds to, in the power amplifier module **100**, the third transmission line transformer **187a** in the combiner **180a**, or the sixth transmission line transformer **187b** in the combiner **180b**.

(84) Thus, in comparison with the power amplifier module **1000** according to the comparative example, the power amplifier module **200** can be reduced in size.

(85) FIG. 9B illustrates the power amplifier module **200** including no transmission line transformer between the antenna switch **282** and the antennas ANT1 and ANT2. As illustrated in FIG. 9B, the combiner **280** includes, for example, the preceding transmission line transformers **281**, the antenna switch **282**, and the combination terminal **284**. The preceding transmission line transformers **281** are connected to the antennas ANT1 and ANT2 sides of the duplexers **250**. Each of the preceding transmission line transformers **281** performs impedance transformation at a predetermined transformation ratio. Specifically, the preceding transmission line transformers **281** correspond to, for example, in the power amplifier module **100**, the seventh and eighth transmission line transformers **182c** and **183c** and the ninth and tenth transmission line transformers **185c** and **186c** in the combiner **180c**, or the eleventh and twelfth transmission line transformers **182d** and **183d** and the thirteenth and fourteenth transmission line transformers **185d** and **186d** in the combiner **180d**. The antenna switch **282** is described above, and thus a description thereof is omitted. The combination terminal **284** is connected, for example, to a path-changing switch **270**. The path-changing switch **270** is similar to the path-changing switch **170**, and thus a description thereof is omitted. Thus, in comparison with the power amplifier module **1000** according to the comparative example, the power amplifier module **200** can be reduced in size.

(86) Summary

(87) The power amplifier module **100** according to the present embodiment includes the first amplifier **110a** that amplifies a power level of an input signal RFin1 (first input signal) in a predetermined frequency band and outputs an amplified signal RFamp1 (first signal) of PC3 (first power level); the first impedance transformer **120a** connected to the first amplifier **110a** and including a transmission line transformer; the second amplifier **110b** that amplifies a power level of an input signal RFin2 (second input signal) in the predetermined frequency band and outputs an amplified signal RFamp2 (second signal) of PC3 (first power level); the second impedance transformer **120b** connected to the second amplifier **110b** and including a transmission line transformer; and the combiner **180** that combines the amplified signal RFamp1 (first signal) inputted through the first impedance transformer **120a** and the amplified signal RFamp2 (second signal) inputted through the second impedance transformer **120b** into an output signal of PC2 (second power level) larger than PC3 (first power level) and includes a transmission line transformer. This enables an increase in transmission efficiency while achieving a reduction in module size.

(88) Furthermore, the power amplifier module **100** according to the present embodiment further includes the antenna switch **160**. The antenna switch **160** includes one terminal (first terminal) of input terminals **161** to which the amplified signal RFamp1 (first signal) is inputted through the first impedance transformer **120a**, one terminal (second terminal) of input terminals **161** to which the

amplified signal RFamp2 (second signal) is inputted through the second impedance transformer **120b**, the first output terminal **162a** that outputs the amplified signal RFamp1 (first signal) to the antenna ANT1 (first antenna), the second output terminal **162b** that outputs the amplified signal RFamp2 (second signal) to the antenna ANT2 (second antenna) different from the antenna ANT1 (first antenna), and the third output terminals **162c** that output the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) to the combiner **180**. This enables an increase in transmission efficiency while achieving a reduction in module size. Furthermore, the combiner **180** is disposed on the antennas ANT1 and ANT2 side of the antenna switch **160**, and thus the impedances at the input and output terminals of the antenna switch **160** are about 50 ohms, therefore making it possible to reduce power loss due to a resistance component of the antenna switch **160**.

(89) Furthermore, the first form of combiner **180a** (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **186a** at which the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) are combined, the first transmission line transformer **182a** that is connected between the combination terminal **186a** and the first input terminal **181a** to which the amplified signal RFamp1 (first signal) is inputted and that performs impedance transformation at a transformation ratio of 1:9 (first transformation ratio) so that an impedance increases, the second transmission line transformer **184a** that is connected between the combination terminal **186a** and the second input terminal **183a** to which the amplified signal RFamp2 (second signal) is inputted and that performs impedance transformation at the transformation ratio of 1:9 (first transformation ratio) so that an impedance increases, the third transmission line transformer **187a** that is connected between the combination terminal **186a** and the output terminal **188a** and that performs impedance transformation at a transformation ratio of 4:1 (second transformation ratio) so that an impedance decreases, and the first isolation section **185a** connected between the first input terminal **181a** and the second input terminal **183a**. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

(90) Furthermore, the second form of combiner **180b** (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **186b** at which the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) are combined, the fourth transmission line transformer **182b** that is connected between the combination terminal **186b** and the third input terminal **181b** to which the amplified signal RFamp1 (first signal) is inputted and that performs impedance transformation at a transformation ratio of 4:1 (third transformation ratio) so that an impedance decreases, the fifth transmission line transformer **184b** that is connected between the combination terminal **186b** and the fourth input terminal **183b** to which the amplified signal RFamp2 (second signal) is inputted and that performs impedance transformation at the transformation ratio of 4:1 (third transformation ratio) so that an impedance decreases, the sixth transmission line transformer **187b** that is connected between the combination terminal **186b** and the output terminal **188b** and that performs impedance transformation at a transformation ratio of 1:9 (fourth transformation ratio) so that an impedance increases, and the second isolation section **185b** connected between the third input terminal **181b** and the fourth input terminal **183b**. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

(91) Furthermore, the third form of combiner **180c** (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **188c** at which the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) are combined, the seventh transmission line transformer **182c** that performs impedance transformation at a transformation ratio of 1:9 (fifth transformation ratio) so that an impedance increases, the eighth transmission line transformer **183c** that is connected in series with the seventh transmission line transformer **182c** through the first connection terminal **1881c** and that performs impedance transformation at a transformation ratio of 4:1 (sixth transformation ratio) so that the impedance decreases, the ninth transmission line transformer **185c** that performs impedance transformation at

the transformation ratio of 1:9 (fifth transformation ratio) so that an impedance increases, the tenth transmission line transformer **186c** that is connected in series with the ninth transmission line transformer **185c** through the second connection terminal **1882c** and that performs impedance transformation at the transformation ratio of 4:1 (sixth transformation ratio) so that the impedance decreases, and the third isolation section **187c**. The seventh transmission line transformer **182c** is connected between the first connection terminal **1881c** and the fifth input terminal **181c** to which the amplified signal RFamp1 (first signal) is inputted. The eighth transmission line transformer **183c** is connected between the first connection terminal **1881c** and the combination terminal **188c**. The ninth transmission line transformer **185c** is connected between the second connection terminal **1882c** and the sixth input terminal **184c** to which the amplified signal RFamp2 (second signal) is inputted. The tenth transmission line transformer **186c** is connected between the second connection terminal **1882c** and the combination terminal **188c**. The third isolation section **187c** is connected between the fifth input terminal **181c** and the sixth input terminal **184c**. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

(92) Furthermore, the fourth form of combiner **180d** (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **188d** at which the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) are combined, the eleventh transmission line transformer **182d** that performs impedance transformation at a transformation ratio of 4:1 (seventh transformation ratio) so that an impedance decreases, the twelfth transmission line transformer **183d** that is connected in series with the eleventh transmission line transformer **182d** through a third connection terminal **1881d** and that performs impedance transformation at a transformation ratio of 1:9 (eighth transformation ratio) so that the impedance increases, the thirteenth transmission line transformer **185d** that performs impedance transformation at the transformation ratio of 4:1 (seventh transformation ratio) so that an impedance decreases, the fourteenth transmission line transformer **186d** that is connected in series with the thirteenth transmission line transformer **185d** through a fourth connection terminal **1882d** and that performs impedance transformation at the transformation ratio of 1:9 (eighth transformation ratio) so that the impedance increases, and the fourth isolation section **187d**. The eleventh transmission line transformer **182d** is connected between the third connection terminal **1881d** and the seventh input terminal **181d** to which the amplified signal RFamp1 (first signal) is inputted. The twelfth transmission line transformer **183d** is connected between the third connection terminal **1881d** and the combination terminal **188d**. The thirteenth transmission line transformer **185d** is connected between the fourth connection terminal **1882d** and the eighth input terminal **184d** to which the amplified signal RFamp2 (second signal) is inputted. The fourteenth transmission line transformer **186d** is connected between the fourth connection terminal **1882d** and the combination terminal **188d**. The fourth isolation section **187d** is connected between the seventh input terminal **181d** and the eighth input terminal **184d**. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

(93) Furthermore, the fifth form of combiner **180e** (combiner) in the power amplifier module **100** according to the present embodiment includes the combination terminal **187e** at which the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) are combined, the fifteenth transmission line transformer **183e** that includes the second end portion **191b** (first end portion) that is connected to the ninth input terminal **181e** and to which the amplified signal RFamp1 (first signal) is inputted, the sixth end portion **193b** (second end portion) that is connected to the tenth input terminal **182e** and to which the amplified signal RFamp2 (second signal) is inputted, and the first end portion **191a** (third end portion) connected to the combination terminal **187e**, and that performs impedance transformation at a transformation ratio of 4:1 (ninth transformation ratio) so that an impedance decreases, the sixteenth transmission line transformer **184e** that is connected in series with the fifteenth transmission line transformer **183e** through the combination terminal **187e** and connected between the combination terminal **187e** and

the output terminal **186e**, and that performs impedance transformation at a transformation ratio of 1:2.25 (tenth transformation ratio) so that the impedance increases, and the fifth isolation section **185e** connected between the ninth input terminal **181e** and the tenth input terminal **182e**. Thus, the low-loss and wideband power amplifier module **100** can be implemented.

(94) Furthermore, the combiner **280** of the power amplifier module **200** according to the present embodiment includes a preceding transmission line transformer **281** (seventeenth transmission line transformer) that is connected to a duplexer **150** (first duplexer) separating a reception signal and the amplified signal RFamp1 (first signal) and that performs impedance transformation at a predetermined transformation ratio (eleventh transformation ratio), a preceding transmission line transformer **281** (eighteenth transmission line transformer) that is connected to a duplexer **150** (second duplexer) separating a reception signal and the amplified signal RFamp2 (second signal) and that performs impedance transformation at the predetermined transformation ratio (eleventh transformation ratio), and the antenna switch **282** including an input terminal (eleventh input terminal) to which the amplified signal RFamp1 (first signal) is inputted through the preceding transmission line transformer **281** (seventeenth transmission line transformer), an input terminal (twelfth input terminal) to which the amplified signal RFamp2 (second signal) is inputted through the preceding transmission line transformer **281** (eighteenth transmission line transformer), a fourth output terminal connected to the antenna ANT1 (first antenna), an output terminal (fifth output terminal) connected to the antenna ANT2 (second antenna) different from the antenna ANT1 (first antenna), and output terminals (sixth output terminals) connected to the combination terminal **284** at which the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) are combined. Thus, the power amplifier module **200** is reduced in size.

(95) Furthermore, the combiner **280** of the power amplifier module **200** according to the present embodiment includes the subsequent transmission line transformer **283** (nineteenth transmission line transformer) that is connected between the combination terminal **284** and an output terminal (for example, a terminal connected to the path-changing switch **270**) and that performs impedance transformation at a predetermined transformation ratio (twelfth transformation ratio). Thus, the power amplifier module **200** can be reduced in size.

(96) Furthermore, the combiner **180** or **280** in the power amplifier module **100** or **200** according to the present embodiment receives input of the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) through the first and second band selection switches **140a** and **140b** that distribute the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) in accordance with the predetermined frequency band. Thus, a module used in a wide band can be implemented.

(97) Furthermore, the combiner **180** in the power amplifier module **100** according to the present embodiment receives input of the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal) through duplexers **150** that distribute the amplified signal RFamp1 (first signal) and the amplified signal RFamp2 (second signal), and a reception signal received. This enables transmission and reception using the same antenna, thus achieving a reduction in module size.

(98) Furthermore, the predetermined frequency band of each input signal RFin in the power amplifier module **100** or **200** according to the present embodiment is a frequency band from 1710 MHz to 2690 MHz. When the first amplifier **110a** and the second amplifier **110b** that support a wide band are used, the number of components can be reduced, thus achieving a reduction in module size.

(99) The above-described embodiments are intended to facilitate understanding of the present disclosure but are not intended for a limited interpretation of the present disclosure. The present disclosure can be changed or improved without departing from the gist thereof and also encompasses equivalents thereof. In other words, appropriate design changes made to the embodiments by those skilled in the art are also encompassed in the scope of the present disclosure.

as long as the changes have features of the present disclosure. The elements included in the embodiments, and the arrangement and so forth of the elements are not necessarily limited to those exemplified herein and can be appropriately changed. While preferred embodiments of the disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the disclosure. The scope of the disclosure, therefore, is to be determined solely by the following claims.

Claims

1. A power amplifier module comprising: a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer; and an antenna switch, wherein the antenna switch includes: a first terminal to which the first signal is inputted through the first impedance transformer, a second terminal to which the second signal is inputted through the second impedance transformer, a first output terminal configured to output the first signal to a first antenna, a second output terminal configured to output the second signal to a second antenna different from the first antenna, and third output terminals configured to output the first signal and the second signal to the combiner.
2. A power amplifier module comprising: a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, a first transmission line transformer connected between the combination terminal and a first input terminal to which the first signal is inputted and configured to perform impedance transformation at a first transformation ratio so that an impedance increases, a second transmission line transformer connected between the combination terminal and a second input terminal to which the second signal is inputted and configured to perform impedance transformation at the first transformation ratio so that an impedance increases, a third transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a second transformation ratio so that an impedance decreases, and a first isolation section connected between the first input terminal and the second input terminal.
3. The power amplifier module according to claim 1, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, a first transmission line transformer connected between the combination terminal and a first input terminal to which the first signal is inputted and configured to perform impedance transformation at a first transformation ratio so that an impedance increases, a second transmission line transformer connected between the combination terminal and a second input terminal to which the second signal is inputted and

configured to perform impedance transformation at the first transformation ratio so that an impedance increases, a third transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a second transformation ratio so that an impedance decreases, and a first isolation section connected between the first input terminal and the second input terminal.

4. A power amplifier module comprising: a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, a fourth transmission line transformer connected between the combination terminal and a third input terminal to which the first signal is inputted and configured to perform impedance transformation at a third transformation ratio so that an impedance decreases, a fifth transmission line transformer connected between the combination terminal and a fourth input terminal to which the second signal is inputted and configured to perform impedance transformation at the third transformation ratio so that an impedance decreases, a sixth transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a fourth transformation ratio so that an impedance increases, and a second isolation section connected between the third input terminal and the fourth input terminal.

5. The power amplifier module according to claim 1, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, a fourth transmission line transformer connected between the combination terminal and a third input terminal to which the first signal is inputted and configured to perform impedance transformation at a third transformation ratio so that an impedance decreases, a fifth transmission line transformer connected between the combination terminal and a fourth input terminal to which the second signal is inputted and configured to perform impedance transformation at the third transformation ratio so that an impedance decreases, a sixth transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a fourth transformation ratio so that an impedance increases, and a second isolation section connected between the third input terminal and the fourth input terminal.

6. A power amplifier module comprising: a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, a seventh transmission line transformer configured to perform impedance transformation at a fifth transformation ratio so that an impedance increases, an eighth transmission line transformer connected in series with the seventh transmission line transformer through a first connection terminal and configured to perform

impedance transformation at a sixth transformation ratio so that the impedance decreases, a ninth transmission line transformer configured to perform impedance transformation at the fifth transformation ratio so that an impedance increases, a tenth transmission line transformer connected in series with the ninth transmission line transformer through a second connection terminal and configured to perform impedance transformation at the sixth transformation ratio so that the impedance decreases, and a third isolation section, wherein the seventh transmission line transformer is connected between the first connection terminal and a fifth input terminal to which the first signal is inputted, wherein the eighth transmission line transformer is connected between the first connection terminal and the combination terminal, wherein the ninth transmission line transformer is connected between the second connection terminal and a sixth input terminal to which the second signal is inputted, wherein the tenth transmission line transformer is connected between the second connection terminal and the combination terminal, and wherein the third isolation section is connected between the fifth input terminal and the sixth input terminal.

7. The power amplifier module according to claim 1, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, a seventh transmission line transformer configured to perform impedance transformation at a fifth transformation ratio so that an impedance increases, an eighth transmission line transformer connected in series with the seventh transmission line transformer through a first connection terminal and configured to perform impedance transformation at a sixth transformation ratio so that the impedance decreases, a ninth transmission line transformer configured to perform impedance transformation at the fifth transformation ratio so that an impedance increases, a tenth transmission line transformer connected in series with the ninth transmission line transformer through a second connection terminal and configured to perform impedance transformation at the sixth transformation ratio so that the impedance decreases, and a third isolation section, wherein the seventh transmission line transformer is connected between the first connection terminal and a fifth input terminal to which the first signal is inputted, wherein the eighth transmission line transformer is connected between the first connection terminal and the combination terminal, wherein the ninth transmission line transformer is connected between the second connection terminal and a sixth input terminal to which the second signal is inputted, wherein the tenth transmission line transformer is connected between the second connection terminal and the combination terminal, and wherein the third isolation section is connected between the fifth input terminal and the sixth input terminal.

8. A power amplifier module comprising: a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, an eleventh transmission line transformer configured to perform impedance transformation at a seventh transformation ratio so that an impedance decreases, a twelfth transmission line transformer connected in series with the eleventh transmission line transformer through a third connection terminal and configured to perform impedance transformation at an eighth transformation ratio so that the impedance increases, a thirteenth transmission line transformer configured to perform impedance transformation at the seventh transformation ratio so that an impedance decreases, a fourteenth transmission line transformer connected in series with the thirteenth transmission line transformer through a fourth connection terminal and configured to perform impedance transformation at the

eighth transformation ratio so that the impedance increases, and a fourth isolation section, wherein the eleventh transmission line transformer is connected between the third connection terminal and a seventh input terminal to which the first signal is inputted, wherein the twelfth transmission line transformer is connected between the third connection terminal and the combination terminal, wherein the thirteenth transmission line transformer is connected between the fourth connection terminal and an eighth input terminal to which the second signal is inputted, wherein the fourteenth transmission line transformer is connected between the fourth connection terminal and the combination terminal, and wherein the fourth isolation section is connected between the seventh input terminal and the eighth input terminal.

9. The power amplifier module according to claim 1, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, an eleventh transmission line transformer configured to perform impedance transformation at a seventh transformation ratio so that an impedance decreases, a twelfth transmission line transformer connected in series with the eleventh transmission line transformer through a third connection terminal and configured to perform impedance transformation at an eighth transformation ratio so that the impedance increases, a thirteenth transmission line transformer configured to perform impedance transformation at the seventh transformation ratio so that an impedance decreases, a fourteenth transmission line transformer connected in series with the thirteenth transmission line transformer through a fourth connection terminal and configured to perform impedance transformation at the eighth transformation ratio so that the impedance increases, and a fourth isolation section, wherein the eleventh transmission line transformer is connected between the third connection terminal and a seventh input terminal to which the first signal is inputted, wherein the twelfth transmission line transformer is connected between the third connection terminal and the combination terminal, wherein the thirteenth transmission line transformer is connected between the fourth connection terminal and an eighth input terminal to which the second signal is inputted, wherein the fourteenth transmission line transformer is connected between the fourth connection terminal and the combination terminal, and wherein the fourth isolation section is connected between the seventh input terminal and the eighth input terminal.

10. A power amplifier module comprising: a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, a fifteenth transmission line transformer including a first end portion that is connected to a ninth input terminal and to which the first signal is inputted, a second end portion that is connected to a tenth input terminal and to which the second signal is inputted, and a third end portion connected to the combination terminal, and configured to perform impedance transformation at a ninth transformation ratio so that an impedance decreases, a sixteenth transmission line transformer connected in series with the fifteenth transmission line transformer through the combination terminal and connected between the combination terminal and an output terminal, and configured to perform impedance transformation at a tenth transformation ratio so that the impedance increases, and a fifth isolation section connected between the ninth input terminal and the tenth input terminal.

11. The power amplifier module according to claim 1, wherein the combiner includes: a combination terminal at which the first signal and the second signal are combined, a fifteenth

transmission line transformer including a first end portion that is connected to a ninth input terminal and to which the first signal is inputted, a second end portion that is connected to a tenth input terminal and to which the second signal is inputted, and a third end portion connected to the combination terminal, and configured to perform impedance transformation at a ninth transformation ratio so that an impedance decreases, a sixteenth transmission line transformer connected in series with the fifteenth transmission line transformer through the combination terminal and connected between the combination terminal and an output terminal, and configured to perform impedance transformation at a tenth transformation ratio so that the impedance increases, and a fifth isolation section connected between the ninth input terminal and the tenth input terminal.

12. A power amplifier module comprising: a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer, wherein the combiner includes; a seventeenth transmission line transformer connected to a first duplexer that separates a reception signal and the first signal and configured to perform impedance transformation at an eleventh transformation ratio, an eighteenth transmission line transformer connected to a second duplexer that separates a reception signal and the second signal and configured to perform impedance transformation at the eleventh transformation ratio, and an antenna switch including an eleventh input terminal to which the first signal is inputted through the seventeenth transmission line transformer, a twelfth input terminal to which the second signal is inputted through the eighteenth transmission line transformer, a fourth output terminal connected to a first antenna, a fifth output terminal connected to a second antenna different from the first antenna, and sixth output terminals connected to a combination terminal at which the first signal and the second signal are combined.

13. The power amplifier module according to claim 12, wherein the combiner includes a nineteenth transmission line transformer connected between the combination terminal and an output terminal and configured to perform impedance transformation at a twelfth transformation ratio.

14. A power amplifier module comprising: a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer, wherein the combiner receives input of the first signal and the second signal through band selection switches configured to distribute the first signal and the second signal in accordance with the predetermined frequency band.

15. A power amplifier module comprising: a first amplifier configured to amplify a power level of a first input signal in a predetermined frequency band and output a first signal of a first power level; a first impedance transformer connected to the first amplifier and including a transmission line transformer; a second amplifier configured to amplify a power level of a second input signal in the predetermined frequency band and output a second signal of the first power level; a second

impedance transformer connected to the second amplifier and including a transmission line transformer; and a combiner configured to combine the first signal inputted through the first impedance transformer and the second signal inputted through the second impedance transformer into an output signal of a second power level larger than the first power level, the combiner including a transmission line transformer, wherein the combiner receives input of the first signal and the second signal through duplexers configured to distribute the first signal and the second signal, and a received reception signal.

16. The power amplifier module according to claim 1, wherein the combiner receives input of the first signal and the second signal through duplexers configured to distribute the first signal and the second signal, and a received reception signal.

17. The power amplifier module according to claim 1, wherein the predetermined frequency band is a frequency band from 1710 MHz to 2690 MHz.

18. The power amplifier module according to claim 2, wherein the predetermined frequency band is a frequency band from 1710 MHz to 2690 MHz.

19. The power amplifier module according to claim 1, wherein the combiner is connected between the antenna switch and the first antenna and the second antenna.
